

Special discrete distributions:

$$\sum_{i=1}^N P(X=i) = \sum_{i=1}^N \frac{1}{N} = 1$$

1. Uniform distribution $X: 1, 2, 3, \dots, N$

$$P(X=i) = \frac{1}{N}, \quad i=1, 2, 3, \dots, N$$

$$E(X) = \sum_{i=1}^N i \cdot \frac{1}{N} = \frac{1}{N} \sum_{i=1}^N i = \frac{1}{N} \cdot \frac{N(N+1)}{2} = \frac{N+1}{2} \checkmark$$

$$E(X^2) = \sum_{i=1}^N i^2 \cdot \frac{1}{N} = \frac{1}{N} \left(\sum_{i=1}^N i^2 \right) = \frac{1}{N} \cdot \frac{N(N+1)(2N+1)}{6} = \frac{(N+1)(2N+1)}{6}$$

$$\therefore V(X) = E(X^2) - (E(X))^2 \checkmark$$

$$= \frac{(N+1)(2N+1)}{6} - \left(\frac{N+1}{2} \right)^2$$

$$= \frac{N+1}{12} \{4N+2-3N-3\}$$

$$= \frac{(N+1)(N-1)}{12}$$

$$= \frac{N^2-1}{12}$$

$$\begin{aligned} E(X^2) &= \sum_x x^2 P(X=x) \\ &= \sum_{i=1}^N i^2 P(X=i) \end{aligned}$$

$$M_X(t) = E(e^{tX})$$

$$= \sum_{i=1}^N e^{ti} P(X=i)$$

$$= \sum_{i=1}^N e^{it} \cdot \frac{1}{N} = \frac{1}{N} \sum_{i=1}^N e^{it} = \frac{1}{N} (e^t + e^{2t} + e^{3t} + \dots + e^{Nt})$$

2. Bernoulli's distribution: $X: 0, 1$

$$P(X=1) = p, \quad P(X=0) = 1-p, \quad 0 < p < 1$$

$$EX = 1 \cdot p + 0 \cdot (1-p) = p$$

$$EX^2 = 1^2 \cdot p + 0^2 \cdot (1-p) = p^2$$

$$\therefore V(X) = EX^2 - (EX)^2$$

$$= p - p^2 = p(1-p)$$

$$= \underline{pq} \text{ where } q = 1-p.$$

3. Binomial Distribution ✓

4. Poisson Distribution

$$P(X=x) = \frac{e^{-\lambda} \lambda^x}{x!}, x = 0, 1, 2, \dots$$

$$M_X(t) = E(e^{tx}) = \sum_{x=0}^{\infty} e^{tx} \cdot P(X=x)$$

$$= \sum_{x=0}^{\infty} e^{tx} \cdot \frac{e^{-\lambda} \lambda^x}{x!}$$

$$= e^{-\lambda} \sum_{x=0}^{\infty} \frac{\lambda^x e^{tx}}{x!}$$

$$= e^{-\lambda} \sum_{x=0}^{\infty} \frac{(\lambda e^t)^x}{x!}$$

$$= e^{-\lambda} \cdot e^{\lambda e^t}$$

$$\boxed{M_X(t) = e^{\lambda(e^t - 1)}}$$

$$M_X(t) = e^{\lambda(e^t - 1)} \cdot \lambda e^t$$

$$M_X''(t) = e^{\lambda(e^t - 1)} \cdot (\lambda e^t)^2 + e^{\lambda(e^t - 1)} \cdot \lambda e^t$$

$$M_X'(0) = \lambda = E(X)$$

$$M_X''(0) = \lambda^2 + \lambda \quad \therefore \text{Var}(X) = E(X^2) - (E(X))^2 = \lambda^2 + \lambda - (\lambda)^2 = \boxed{\lambda}$$

$$e^a = 1 + a + \frac{a^2}{2!} + \frac{a^3}{3!} + \dots$$

$$\boxed{e^a = \sum_{x=0}^{\infty} \frac{a^x}{x!}}$$

$$\text{Bin}(2000, \frac{.99}{\sqrt{.00}})$$

$$P = \binom{2000}{40}$$

$$P(X=40) =$$

$$\lim_{n \rightarrow \infty} \left(1 + \frac{a}{n}\right)^n = e^a$$

MGF: $X \sim \text{Bin}(n, p)$ $\binom{np = \lambda}{\Rightarrow p = \frac{\lambda}{n}}$

$$M_X(t) = (pe^t + q)^n, \quad q = 1 - p$$

$$= \left(\frac{\lambda}{n} e^t + 1 - \frac{\lambda}{n}\right)^n$$

$$= \left(1 + \frac{\lambda(e^t - 1)}{n}\right)^n$$

$$\rightarrow e^{\lambda(e^t - 1)} \quad \text{as } n \rightarrow \infty$$

$\therefore M_X(t) = e^{\lambda(e^t - 1)}$ is the moment generating function of a Poisson distribution $P(\lambda)$, since MGF uniquely determines the distribution. $X \sim P(\lambda)$.

$$M_X(t) = (pe^t + q)^n$$

$$M_X(t) = \left(\frac{2}{3}e^t + \frac{1}{3}\right)^4$$

$$X \sim \text{Bin}(4, \frac{2}{3}) \quad P(X=2) = \binom{4}{2} \left(\frac{2}{3}\right)^2 \left(\frac{1}{3}\right)^2$$

5. Geometric distribution:-

Suppose independent Bernoullian trials are conducted till a success is achieved. Let X represents no of trials needed for the first success.

$$X: 1, 2, 3, 4, 5, \dots$$

$$0 < p < 1$$

$$q = 1 - p$$

$$\underbrace{f f f \dots f}_j \cdot \frac{1}{f}$$

$$P(X=j) = q^{j-1} p, \quad j=1, 2, 3, \dots$$

$$\sum_{j=1}^{\infty} P(X=j) = \sum_{j=1}^{\infty} q^{j-1} p$$

$$= p \sum_{j=1}^{\infty} q^{j-1}$$

$$= p (1 + q + q^2 + \dots), \quad 0 < q < 1$$

$$= p \cdot \frac{1}{1-q} = 1$$

$$0 < q < 1$$

$$|x| < 1$$

$$\frac{1}{1-x} = 1 + x + x^2 + x^3 + \dots$$

$$(1-x)^{-2} = 1 + 2x + 3x^2 + \dots$$

$$|x| < 1$$

$$E(X) = \sum_{j=1}^{\infty} j P(X=j)$$

$$= \sum_{j=1}^{\infty} j \cdot q^{j-1} p = p \sum_{j=1}^{\infty} j q^{j-1} = p \cdot \frac{1}{(1-q)^2} = p \cdot \frac{1}{p^2}$$

$$\boxed{E(X) = \frac{1}{p}}$$

$$E(X^2) = \frac{q+1}{p^2}$$

$$V(X) = E(X^2) - (E(X))^2$$

$$= \frac{q+1}{p^2} - \frac{1}{p^2} = \frac{q}{p^2}$$

$$\begin{aligned} M_X(t) &= \sum_{j=1}^{\infty} e^{tj} q^{j-1} p \\ &= p e^t \sum_{j=1}^{\infty} (q e^t)^{j-1} \end{aligned}$$

$$M_X(t) = \frac{p e^t}{1 - q e^t}, \quad \text{provided } 0 < q e^t < 1$$

Ex: Suppose independent tests are conducted on monkeys to develop a vaccine. If the probability of success is $\frac{1}{3}$ in each trial, what is the probability that at least 5 trials are needed to get the first success?

Sol: Let X denote the number of trials needed.

$$P_X(j) = P(X=j) = \left(\frac{2}{3}\right)^{j-1} \frac{1}{3}, \quad j=1, 2, 3, \dots$$

$$\begin{aligned} P(X \geq 5) &= \sum_{j=5}^{\infty} P_X(j) = \sum_{j=5}^{\infty} \left(\frac{2}{3}\right)^{j-1} \frac{1}{3} \\ &= \left(\frac{2}{3}\right)^4 \cdot \frac{1}{3} \left(1 + \frac{2}{3} + \left(\frac{2}{3}\right)^2 + \left(\frac{2}{3}\right)^3 + \dots\right) \\ &= \left(\frac{2}{3}\right)^4 \cdot \frac{1}{3} \cdot \frac{1}{1 - \frac{2}{3}} \\ &= \left(\frac{2}{3}\right)^4 \end{aligned}$$

$$\square \quad X \sim \text{Geo}(p) \quad P(X=j) = q^{j-1} p, \quad j=1, 2, 3, \dots$$

$$\underline{P(X > m)} = \sum_{j=m+1}^{\infty} q^{j-1} p = q^m p (1 + q + q^2 + \dots)$$

$$= q^m p \cdot \frac{1}{1-q} = q^m p \cdot \frac{1}{p} = \underline{q^m}$$

$$P(\underbrace{X > m+n}_A | \underbrace{X > n}_B) = \frac{P(A \cap B)}{P(B)} = \frac{P(X > m+n, X > n)}{P(X > n)}$$

$$= \frac{P(X > m+n)}{P(X > n)} = \frac{q^{m+n}}{q^n} = q^m = P(X > m)$$

$P(X > m+n | X > n) = P(X > m)$
 ↙
 Memoryless property of Geometric distribution.